lagCAMB

Beyond-ΛCDM Models for CAMB

User Manual

Dark Matter Interactions • Modified Gravity
Interacting Dark Energy • Ultralight Axions
Warm/Decaying Dark Matter

Version 1.0 — May 16, 2026

Based on CAMB (Code for Anisotropies in the Microwave Background)

A. Lewis, A. Challinor & A. Lasenby (2000)

Contents

1 Introduction

lagCAMB extends the standard CAMB Boltzmann solver with a modular framework for beyond-ΛCDM physics. All new models are implemented as plug-in classes inheriting from `TDarkMatterModel` (dark matter sector) or `TDarkEnergyModel` (dark energy / modified gravity), with minimal modifications to the core solver.

1.1 Quick Start

All models follow the same pattern:

```
import camb
from camb.dark_matter import DMBaryonScattering  # or any model

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)

# Activate a dark matter model
pars.DarkMatter = DMBaryonScattering()
pars.DarkMatter.set_params(sigma_dmb=1e-41, n_dmb=0, m_dm=1.0)

results = camb.get_results(pars)
cls = results.get_cmb_power_spectra(CMB_unit='muK')['total']
```

1.2 Cosmological Baseline

Unless otherwise noted, all comparison plots use the following baseline cosmology:

Parameter	Value	Description
H_0	67.5 km/s/Mpc	Hubble constant
$\Omega_b h^2$	0.022	Baryon density
$\Omega_c h^2$	0.122	CDM density
A_s	2.1×10^{-9}	Scalar amplitude
n_s	0.965	Spectral index
$\ell_{\max}$	2500	Multipole truncation

1.3 Model Summary

Model	**Type**	**Primary Effect**	**Reference**
DM–Baryon	DM	CDM–baryon drag	Boddy & Gluscevic 2018
DM–DR (ETHOS)	DM	DM–dark radiation coupling	Cyr-Racine+ 2016
Decaying DM	DM	CDM decay to daughter + DR	Blackadder+ 2014
DM–Neutrino	DM	CDM–neutrino scattering	He+ 2023
Fuzzy DM	DM	Quantum Jeans suppression	Hu+ 2000
Warm DM	DM	Free-streaming suppression	Viel+ 2005
DM–Photon	DM	CDM–photon opacity	Boddy & Gluscevic 2018
Multi-IDM	DM	Combined channels	Becker+ 2021
ETHOS Transfer (Murgia)	DM	Small-scale $T(k)$ fit	Murgia+ 2017
ETHOS Transfer (Physical)	DM	ETHOS params $\to T(k)$	Cyr-Racine+ 2016
Interacting DE	DE	DM–DE energy exchange	Costa+ 2017
Horndeski	DE/MG	Scalar-tensor gravity (QSA)	Bellini & Sawicki 2014
μ-Σ MG	MG	Phenomenological MG	Zhao+ 2009

2 Dark Matter Models

2.1 DM–Baryon Scattering

Physics

Dark matter scatters off baryons with a velocity-dependent momentum-transfer cross section:

$$\sigma_{\rm MT} = \sigma_0 \left(\frac{v}{c}\right)^n \tag{1}$$

where $n \in \{-4, -2, 0, 2, 4\}$. This adds a drag force between the CDM and baryon fluids, suppressing small-scale structure and modifying the CMB damping tail. The back-reaction on baryons is included via momentum conservation.

Parameters

Name	Type	Default	Description
`sigma_dmb`	float	0.0	Cross section σ_0 [cm^2]
`n_dmb`	int	0	Velocity index $(-4, -2, 0, 2, 4)$
`m_dm`	float	100.0	DM particle mass [GeV]

```python
from camb.dark_matter import DMBaryonScattering

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkMatter = DMBaryonScattering()
pars.DarkMatter.set_params(sigma_dmb=1e-41, n_dmb=0, m_dm=1.0)
# n_dmb=0: constant cross section
# Lighter DM (m_dm=1 GeV) gives stronger effect

results = camb.get_results(pars)
```

DM-Baryon Scattering

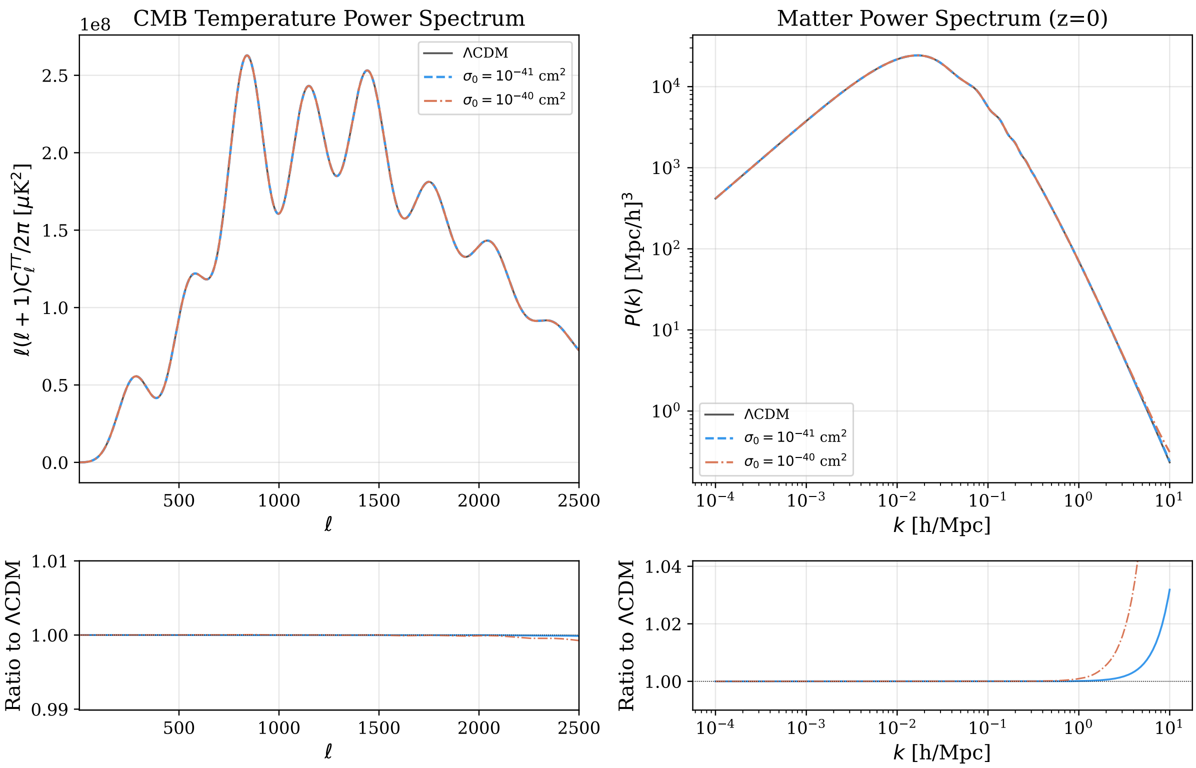

Figure 1: DM–Baryon scattering: CMB TT power spectrum and matter power spectrum compared to ΛCDM. Parameters: $n = 0$ (constant cross section), $m_{\rm DM} = 1$ GeV.

Note

Drag rates are internally capped at $10^4 \times \dot{a}/a$ to prevent ODE stiffness at very early times. For $n_{\rm dmb} = -4$ (Coulomb-like), the effect is strongest at low velocities (late times).

2.2 DM–Dark Radiation (ETHOS)

Physics

In the ETHOS (Effective Theory of Structure Formation) framework, dark matter couples to a dark radiation sector via scattering. DM evolves as a fluid (δ_{DM}, v_{DM}) while DR is described by a Boltzmann hierarchy ($F_0, F_1, \ldots, F_{\ell_{\max}}$). The opacity is parameterized as:

$$\dot{\kappa} = \sum_{n=2}^{6} a_n \left(\frac{T_{\mathrm{DR}}}{T_0}\right)^n \tag{2}$$

The interaction produces dark acoustic oscillations (DAO) analogous to baryon acoustic oscillations. Three regimes are automatically handled: tight coupling (TCA, $\Gamma \gg k$), full Boltzmann hierarchy, and ultra-relativistic fluid approximation (UFA, $k\tau \gg 1$).

Parameters

Name	Type	Default	Description
`omdmdrh2`	float	0.0	Interacting DM density $\Omega_{\mathrm{dmdr}}h^2$
`N_dark`	float	0.0	DR effective number ΔN_{eff}
`a_dark`	list	None	ETHOS opacity coefficients $[a_2, \ldots, a_6]$
`alpha_l_dark`	float	1.0	Angular damping for $\ell \geq 2$
`lmax_dr`	int	15	DR hierarchy truncation
`cs2_dm`	float	0.0	DM sound speed squared

```python
from camb.dark_matter import DMDR_ETHOS

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkMatter = DMDR_ETHOS()
pars.DarkMatter.set_params(
    omdmdrh2=0.005,        # 5% of CDM is interacting
    N_dark=1.0,            # 1 DR species
    a_dark=[1e5],          # strong coupling
    lmax_dr=15             # Boltzmann hierarchy depth
)
results = camb.get_results(pars)
```

DM-Dark Radiation (ETHOS)

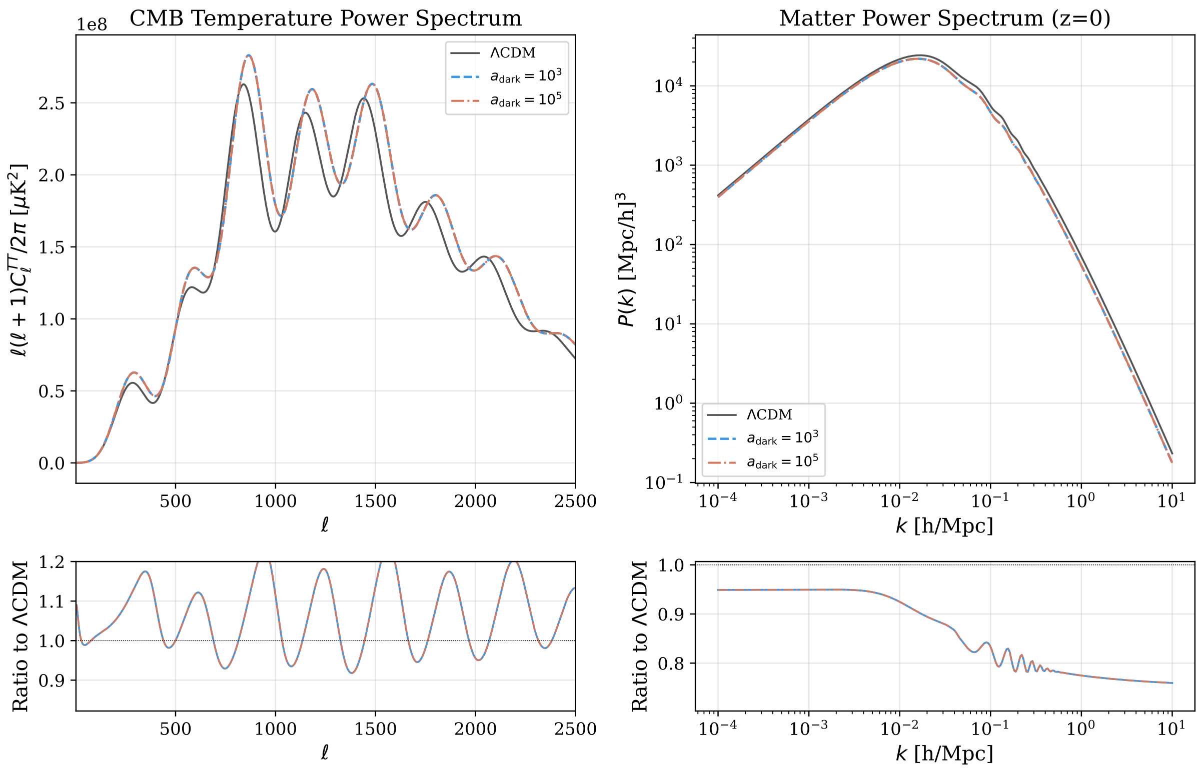

Figure 2: DM–DR (ETHOS): Dark acoustic oscillations appear in both CMB and $P(k)$ at scales entering the horizon before DM–DR decoupling. Stronger coupling ($a_{\rm dark} = 10^5$) produces deeper oscillations.

Note

The implementation uses a tight-coupling approximation (TCA) when the DM–DR interaction rate exceeds $50k$, eliminating stiff coupling terms from the ODE system. This allows stable integration even for very large $a_{\rm dark}$ values ($\sim 10^7$).

2.3 Decaying Dark Matter

Physics

A fraction $f_{\rm dcdm}$ of CDM decays into a massive daughter particle and dark radiation:

$$\mathrm{DM}_{\rm parent} \to \mathrm{DM}_{\rm daughter}\ (\text{mass fraction } \epsilon) + \mathrm{DR} \tag{3}$$

The parent density evolves as $\rho_{\rm dcdm}(a) = \rho_0\, a^{-3}\, e^{-\Gamma t(a)}$. The daughter receives a velocity kick $v_{\rm kick} = (1-\epsilon^2)/(2\epsilon)$, and the DR is accumulated from decay products evolving via a truncated Boltzmann hierarchy.

Parameters

Name	Type	Default	Description
Gamma_dcdm	float	0.0	Decay rate [km/s/Mpc]
epsilon_dcdm	float	1.0	$m_{\rm daughter}/m_{\rm parent}$
f_dcdm	float	1.0	Fraction of CDM that decays

```python
from camb.dark_matter import DecayingDM

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkMatter = DecayingDM()
pars.DarkMatter.set_params(
    Gamma_dcdm=100,        # decay rate ~ H_0
    epsilon_dcdm=0.9,      # 10% mass lost to DR per decay
    f_dcdm=0.1             # 10% of CDM decays
)
results = camb.get_results(pars)
```

Decaying Dark Matter

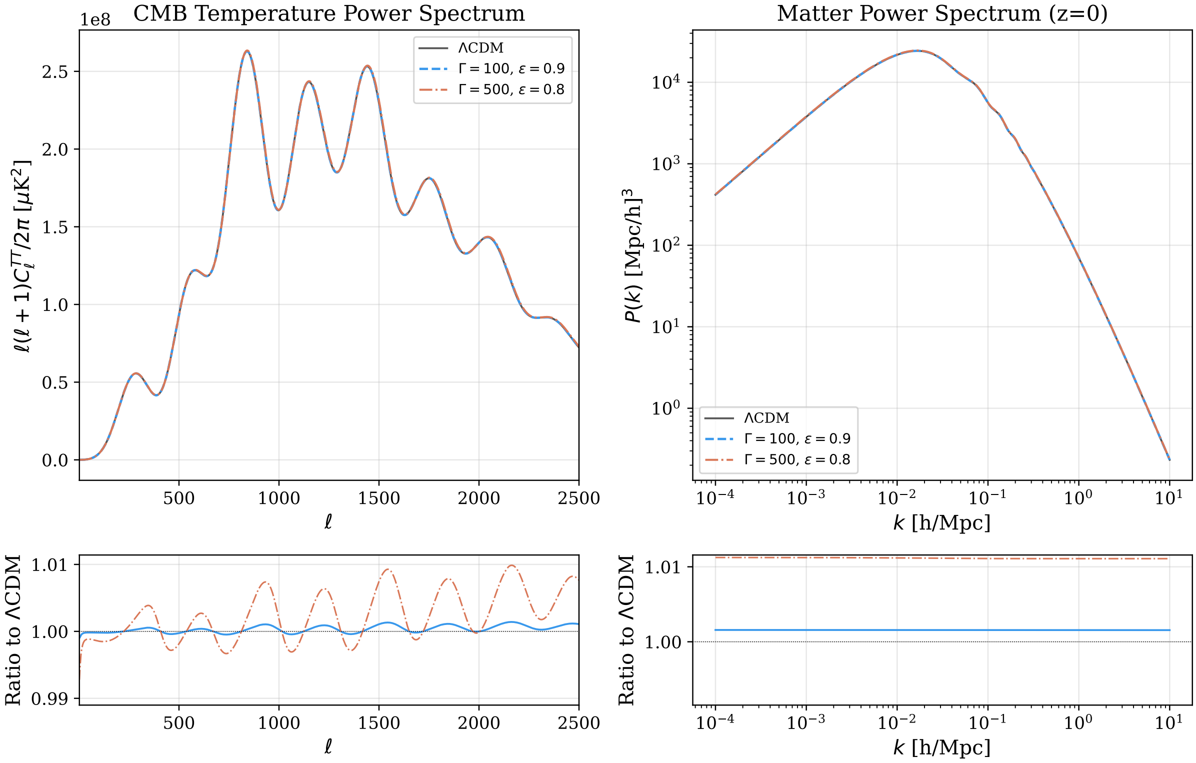

Figure 3: Decaying DM: Decay transfers CDM energy to relativistic DR, reducing matter density at late times. The daughter particle's velocity kick also contributes to small-scale suppression.

Note

The decay rate Γ is in the same units as H_0 [km/s/Mpc]. $\Gamma \sim H_0 \approx 67.5$ means the decay lifetime is comparable to the age of the universe. $\epsilon = 1$ corresponds to no mass loss (no decay effect).

2.4 DM–Neutrino Scattering

Physics

Dark matter scatters off neutrinos with an energy-dependent cross section:

$$\sigma = \sigma_0 \left(\frac{E_\nu}{1 \text{ MeV}} \right)^n \tag{4}$$

This adds drag to the CDM velocity equation and collision damping to the neutrino Boltzmann hierarchy, coupling CDM to the neutrino free-streaming scale.

Parameters

Name	Type	Default	Description
`sigma_dmnu`	float	0.0	Cross section at $E_0 = 1$ MeV [cm^2]
`n_dmnu`	int	0	Energy index (0 or 2)
`m_dm`	float	100.0	DM mass [GeV]

```python
from camb.dark_matter import DMNeutrinoScattering

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkMatter = DMNeutrinoScattering()
pars.DarkMatter.set_params(
    sigma_dmnu=1e-33,      # cross section [cm^2]
    n_dmnu=0,              # constant cross section
    m_dm=0.1               # 100 MeV DM
)
results = camb.get_results(pars)
```

DM-Neutrino Scattering

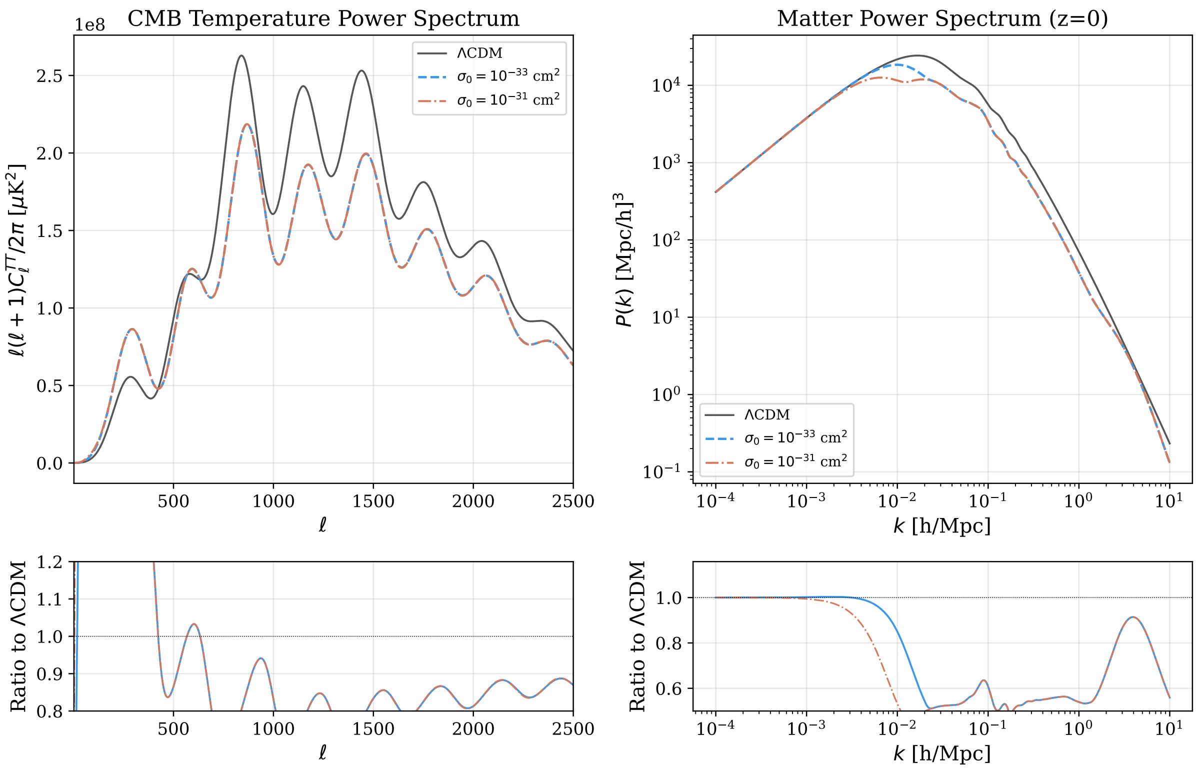

Figure 4: DM–Neutrino scattering: Coupling CDM to neutrinos delays matter domination on small scales and suppresses the matter power spectrum. Lighter DM gives a stronger per-particle scattering rate.

2.5 Fuzzy Dark Matter

Physics

Ultralight axion dark matter ($m \sim 10^{-22}$ eV) with de Broglie wavelength on astrophysical scales. The effective fluid has a scale-dependent sound speed:

$$c_s^2(k,a) = \frac{k^2}{4\,m^2\,a^2} \tag{5}$$

producing a quantum Jeans scale $k_J^2 = 4\,m_{\rm conf}\,a^2\,\dot{a}/a$ below which density perturbations are suppressed by quantum pressure.

Parameters

Name	Type	Default	Description
`m_axion`	float	10^{-22}	Axion mass [eV]
`omega_axion_h2`	float	0.0	Axion density $\Omega_a h^2$ (0 uses `f_axion`)
`f_axion`	float	1.0	Fraction of CDM that is ULDM

```
from camb.dark_matter import FuzzyDM

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkMatter = FuzzyDM()
pars.DarkMatter.set_params(
    m_axion=1e-24,          # 10^{-24} eV
    f_axion=0.3             # 30% of CDM is ULDM
)
results = camb.get_results(pars)
```

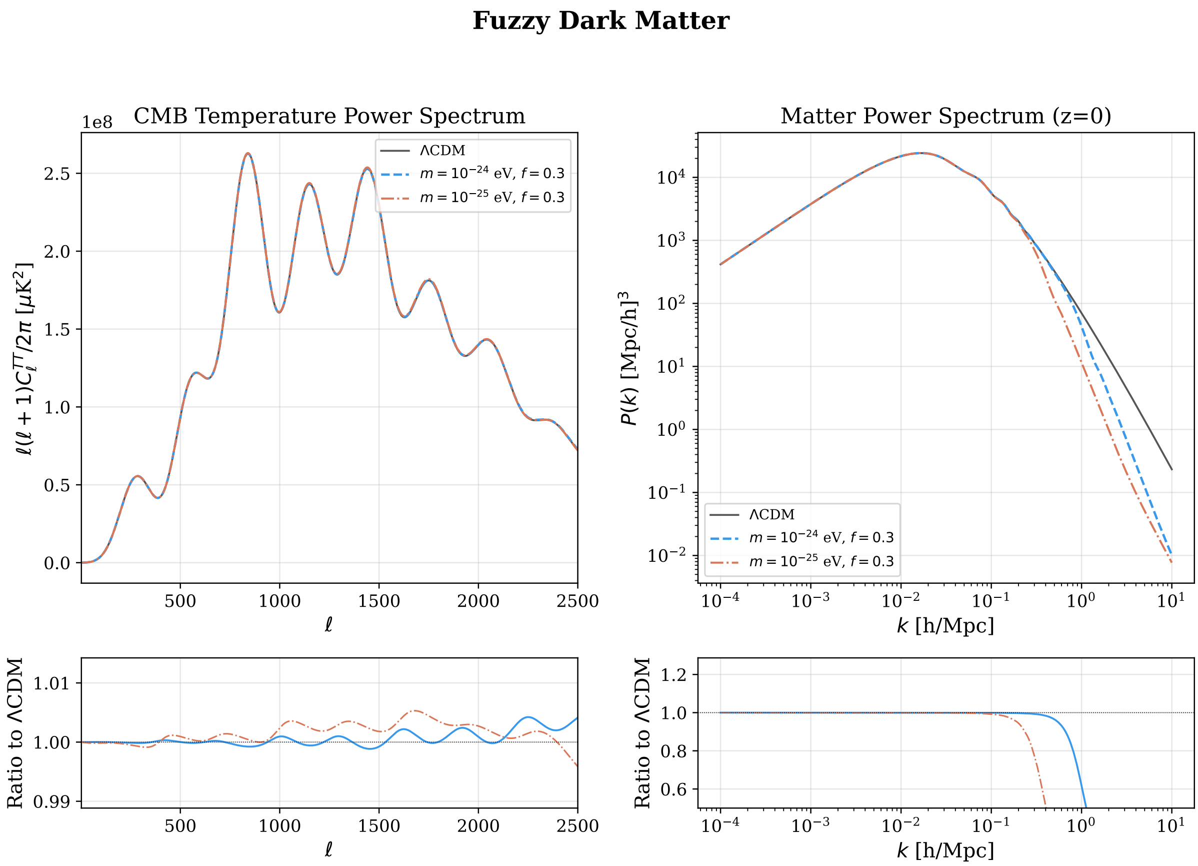

Figure 5: Fuzzy DM: Quantum pressure suppresses $P(k)$ below the Jeans scale, with smaller masses producing suppression at larger scales. The CMB is affected through the integrated Sachs–Wolfe effect.

Note

A full Klein–Gordon field implementation (`FuzzyDMField`) is also available in the DarkEnergy slot, solving the exact background evolution including the frozen-field era and Passaglia–Hu improved EFA for perturbations. See Section 3.4.

2.6 Warm Dark Matter

Physics

Thermal relic WDM with free-streaming that suppresses small-scale structure. In transfer-function mode (default):

$$T(k) = \left[1 + (\alpha\, k)^{2\nu}\right]^{-5/\nu} \tag{6}$$

where α depends on m_{WDM}, Ω_{WDM}, and h. In Boltzmann mode, WDM is treated as an extra massive neutrino species using CAMB's full neutrino hierarchy.

Parameters

Name	Type	Default	Description
`m_wdm`	float	3.0	WDM mass [keV]
`T_wdm_ratio`	float	1.0	T_{WDM}/T_ν ratio
`Omega_wdm_h2`	float	0.0	WDM density (0 = all CDM)
`boltzmann_mode`	bool	False	Full Boltzmann vs transfer function

```python
from camb.dark_matter import WarmDM

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkMatter = WarmDM()
pars.DarkMatter.set_params(m_wdm=1.0)  # 1 keV WDM
results = camb.get_results(pars)
```

Warm Dark Matter

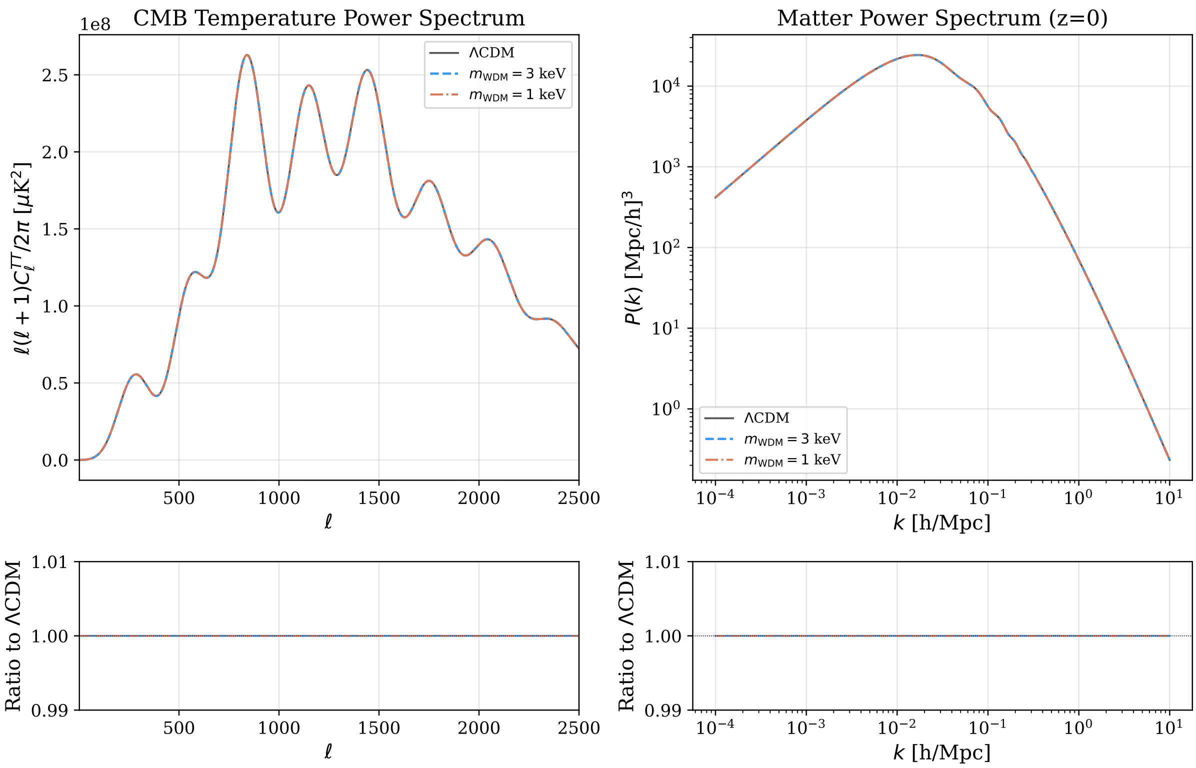

Figure 6: Warm DM: Free-streaming suppresses $P(k)$ at scales below the free-streaming horizon. Lighter particles ($m = 1$ keV) produce stronger suppression at larger scales.

Note

In transfer-function mode (default), the CMB power spectrum is unchanged from ΛCDM. Only the matter power spectrum is modified via the transfer function. Set `boltzmann_mode=True` to include CMB effects via the full Boltzmann hierarchy.

2.7 DM–Photon Scattering

Physics

DM scatters off photons with a temperature-dependent cross section:

$$\sigma = \sigma_{\rm Th}\, u_{\text{idm-g}}\, \frac{m_{\rm DM}}{100\ \text{GeV}} \left(\frac{T}{T_0}\right)^{n_{\text{idm-g}}} \tag{7}$$

This adds opacity to the photon hierarchy (extra scattering beyond Thomson) and drag to the CDM velocity.

Parameters

Name	Type	Default	Description
`u_idm_g`	float	0.0	Dimensionless coupling
`n_idm_g`	int	0	Temperature power index
`m_dm`	float	100.0	DM mass [GeV]

```python
from camb.dark_matter import DMPhotonScattering

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkMatter = DMPhotonScattering()
pars.DarkMatter.set_params(u_idm_g=1e-4, n_idm_g=0, m_dm=100.)
results = camb.get_results(pars)
```

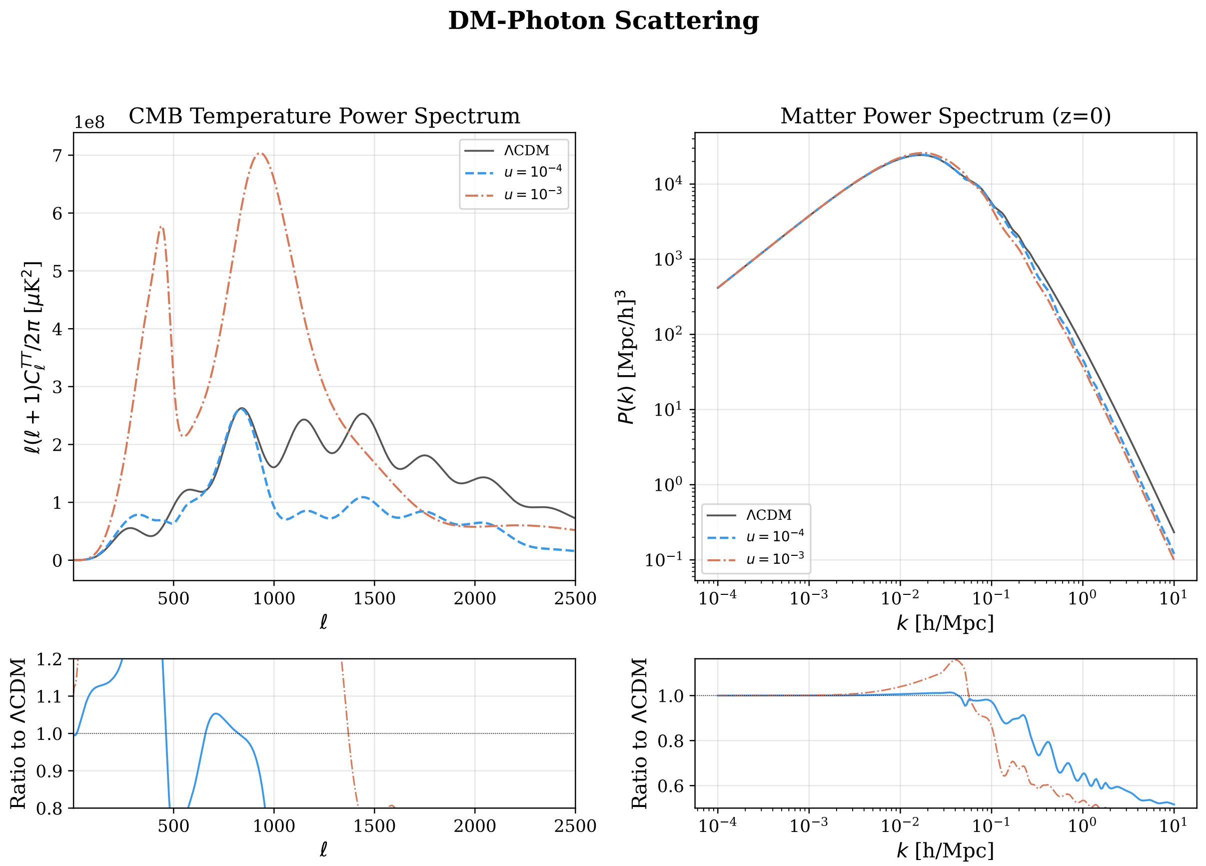

Figure 7: DM–Photon scattering: Extra photon opacity beyond Thomson scattering modifies Silk damping and suppresses small-scale power. Stronger coupling ($u = 10^{-3}$) shows pronounced CMB damping tail changes.

2.8 Multi-Channel Interacting DM

Physics

A single DM fluid with up to three simultaneous interaction channels:

- **Dark Radiation**: ETHOS-style DM–DR Boltzmann hierarchy
- **Baryon**: momentum-transfer scattering
- **Photon**: Thomson-like opacity

Channels activate independently based on parameter values, allowing combined studies of multiple DM interactions.

Parameters

All parameters from DMDR_ETHOS, DMBaryon, and DMPhoton are available:

Channel	Parameters
DR	`omdmdrh2, N_dark, a_dark, alpha_l_dark, lmax_dr`
Baryon	`sigma_dmb, n_dmb`
Photon	`u_idm_g, n_idm_g`
Common	`m_dm, cs2_dm`

```python
from camb.dark_matter import MultiInteractingDM

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkMatter = MultiInteractingDM()
pars.DarkMatter.set_params(
    # DR channel
    omdmdrh2=0.005, N_dark=1.0, a_dark=[1e4],
    # Baryon channel
    sigma_dmb=1e-42, n_dmb=0,
    # Common
    m_dm=10.0
)
results = camb.get_results(pars)
```

Multi-Channel Interacting DM

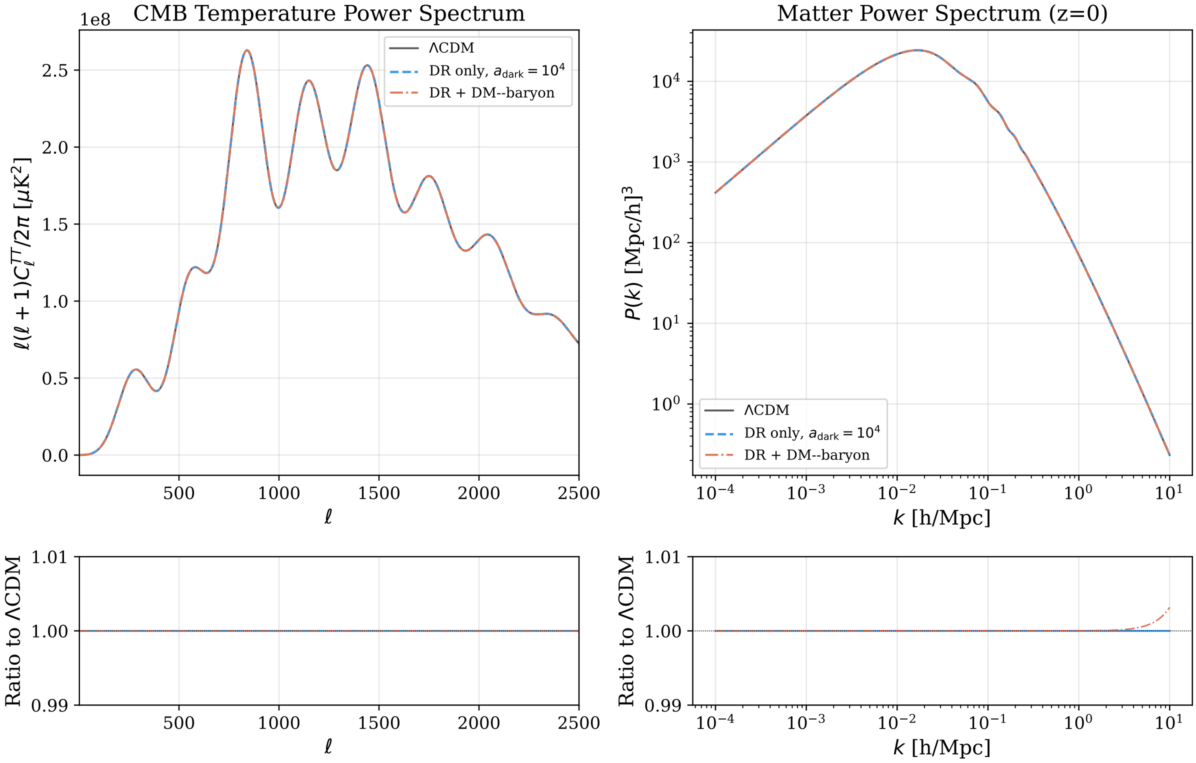

Figure 8: Multi-Channel IDM: combined DR (ETHOS) and DM–baryon channels. Adding a baryon-scattering channel on top of DR coupling produces additional small-scale suppression beyond the DAO bumps from the DR sector alone.

Note

Each channel contributes independently to the DM drag force. The number of perturbation equations varies dynamically: 1 (velocity only) to $1 + 1 + (\ell_{\rm max,DR} + 1)$ when DR is active.

2.9 ETHOS Transfer (Murgia α–β–γ fit)

Physics

A fast analytic surrogate for ETHOS small-scale suppression of the matter power spectrum. The transfer function follows the Murgia et al. (2017) three-parameter fit:

$$T(k) = \left[1 + (\alpha\, k)^{\beta}\right]^{\gamma} \tag{8}$$

applied directly to $P(k) \to T(k)^2 P(k)$ and $\sigma_R \to T(k)\sigma_R$. The defaults $(\beta, \gamma) = (2.24, -4.46)$ match Murgia+ 2017 best-fit values; α [Mpc/h] sets the suppression scale. This model does NOT modify the CMB (no DR sector, no perturbations).

Parameters

Name	Type	Default	Description
`alpha_mpch`	float	0.0	Suppression scale α [Mpc/h] ($0 = \Lambda$CDM)
`beta_mur`	float	2.24	Murgia exponent β
`gamma_mur`	float	-4.46	Murgia exponent γ

```
from camb.dark_matter import ETHOSTransferMurgia

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkMatter = ETHOSTransferMurgia()
pars.DarkMatter.set_params(alpha_mpch=0.05, beta_mur=2.24, gamma_mur
    =-4.46)
results = camb.get_results(pars)
```

ETHOS Transfer (Murgia)

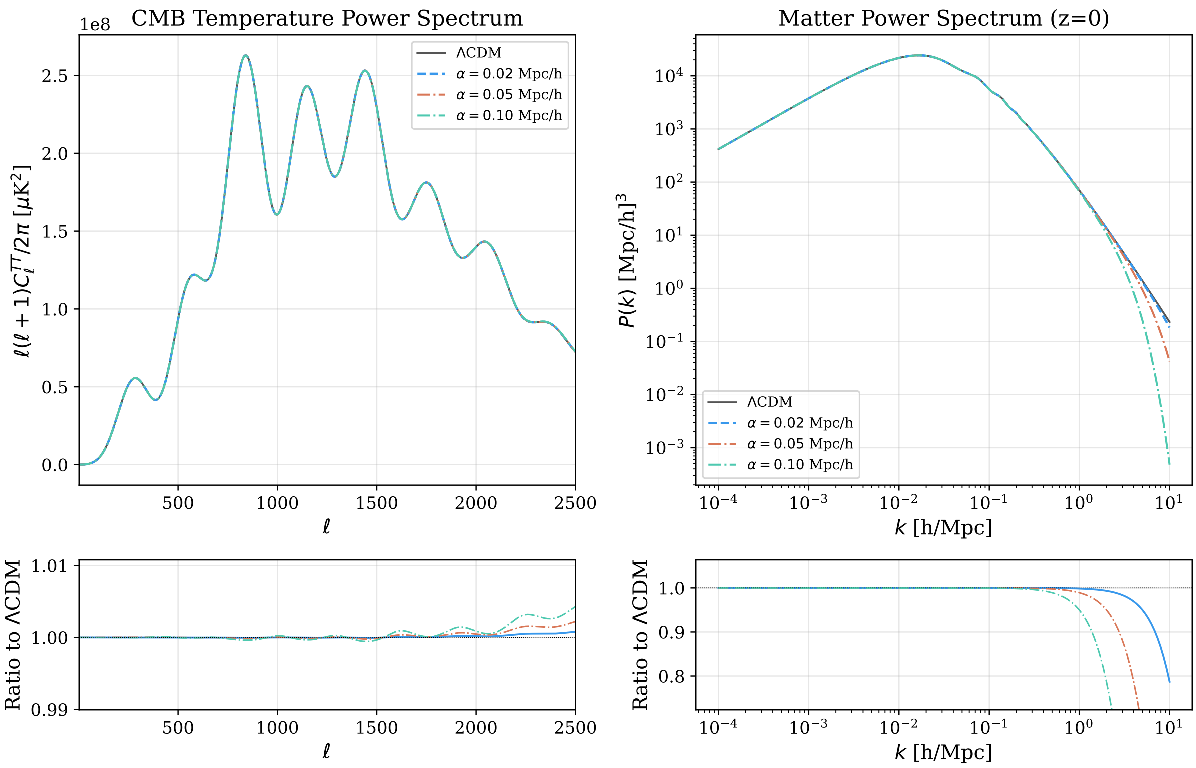

Figure 9: ETHOS Transfer (Murgia): The α–β–γ fitting form imposes a sharp small-scale cutoff in $P(k)$. Larger α moves the cutoff to larger scales. CMB is unchanged because the model only post-processes the matter transfer function.

Note

This is a fast surrogate for parameter scans / Lyman-α-style constraints; for the full self-consistent ETHOS solution (DAO + CMB effects) use `DMDR_ETHOS` (Section 2.2). For physical ETHOS parameters ($a_{\rm dark}, \xi_{\rm DR}, \ldots$) mapped to α, see `ETHOSTransferPhysical` below.

2.10 ETHOS Transfer (Physical surrogate)

Physics

Same Murgia $T(k)$ form as above, but with α derived from physical ETHOS parameters via a power-law fit:

$$\alpha\ [\mathrm{Mpc/h}] = A_0 \left(\frac{a_{\mathrm{dark,n}}}{a_{\mathrm{ref}}}\right)^{p_a} \left(\frac{\xi_{\mathrm{DR}}}{\xi_{\mathrm{ref}}}\right)^{p_\xi} \left(\frac{\omega_{\mathrm{dmdr}}}{0.12}\right)^{p_\omega} \tag{9}$$

with default $(A_0, p_a, p_\xi, p_\omega) = (0.5,\ 0.25,\ 1.0,\ -0.1)$ giving an order-of-magnitude correct mapping for $n_{\mathrm{dark}} = 2$ (dark Thomson-like opacity). The fit constants are user-exposed so each science application can recalibrate against the full DMDR_ETHOS solution for its parameter ranges.

Parameters

Name	Type	Default	Description
`a_dark_n`	float	0.0	ETHOS opacity coefficient [Mpc^{-1}]
`xi_dr`	float	0.5	T_{DR}/T_γ at recombination
`omdmdrh2`	float	0.12	Interacting DM density $\Omega_{\mathrm{dmdr}}h^2$
`n_dark`	int	2	Opacity index (advisory)
`A0_fit, a_ref, xi_ref`	float	0.5, 1e-4, 0.5	Fit reference point
`p_a, p_xi, p_om`	float	0.25, 1.0, -0.1	Fit exponents
`beta_mur, gamma_mur`	float	2.24, -4.46	Murgia exponents

```python
from camb.dark_matter import ETHOSTransferPhysical

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkMatter = ETHOSTransferPhysical()
pars.DarkMatter.set_params(a_dark_n=4e-4, xi_dr=0.5, omdmdrh2=0.12)
results = camb.get_results(pars)
```

ETHOS Transfer (Physical)

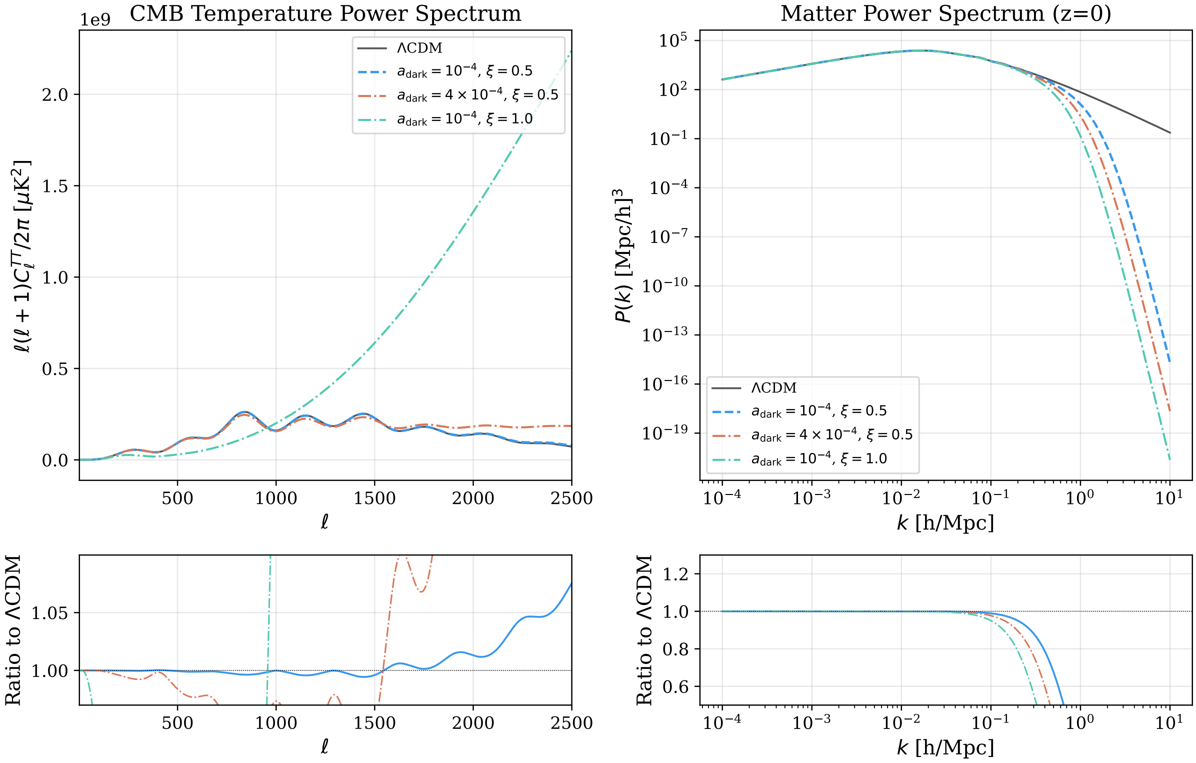

Figure 10: ETHOS Transfer (Physical): Increasing the ETHOS opacity $a_{\rm dark,n}$ or the dark-radiation temperature ratio $\xi_{\rm DR}$ pushes the suppression scale to larger scales. The default power-law mapping is intentionally a rough fit; recalibrate the $A_0, p_a, p_\xi, p_\omega$ exponents against `DMDR_ETHOS` for your target parameter range.

Note

This is a parameter-scan surrogate. It does NOT reproduce dark acoustic oscillations or CMB effects — use `DMDR_ETHOS` (Section 2.2) for the full self-consistent calculation.

3 Dark Energy & Modified Gravity Models

3.1 Interacting Dark Energy

Physics

Dark matter and dark energy exchange energy-momentum Q^μ. The background conservation equations become:

$$\dot{\rho}_c + 3H\rho_c = Q, \qquad \dot{\rho}_{\rm de} + 3(1+w)H\rho_{\rm de} = -Q \tag{10}$$

Three interaction kernels:
1. $Q = \xi\, H\, \rho_{\rm de}$
2. $Q = \xi\, H\, \rho_c$
3. $Q = \xi\, H\, (\rho_c + \rho_{\rm de})$

Perturbation equations are derived consistently from the energy-momentum tensor.

Parameters

Name	Type	Default	Description
`w`	float	-1.0	$w(0)$ equation of state
`wa`	float	0	$-dw/da(0)$
`xi_ide`	float	0.0	Coupling strength
`interaction_type`	int	1	Kernel type (1, 2, or 3)
`cs2_ide`	float	1.0	DE sound speed squared

```python
from camb.dark_energy import InteractingDE

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkEnergy = InteractingDE()
pars.DarkEnergy.set_params(
    w=-0.9,
    xi_ide=0.05,
    interaction_type=1    # Q = xi * H * rho_de
)
results = camb.get_results(pars)
```

Interacting Dark Energy

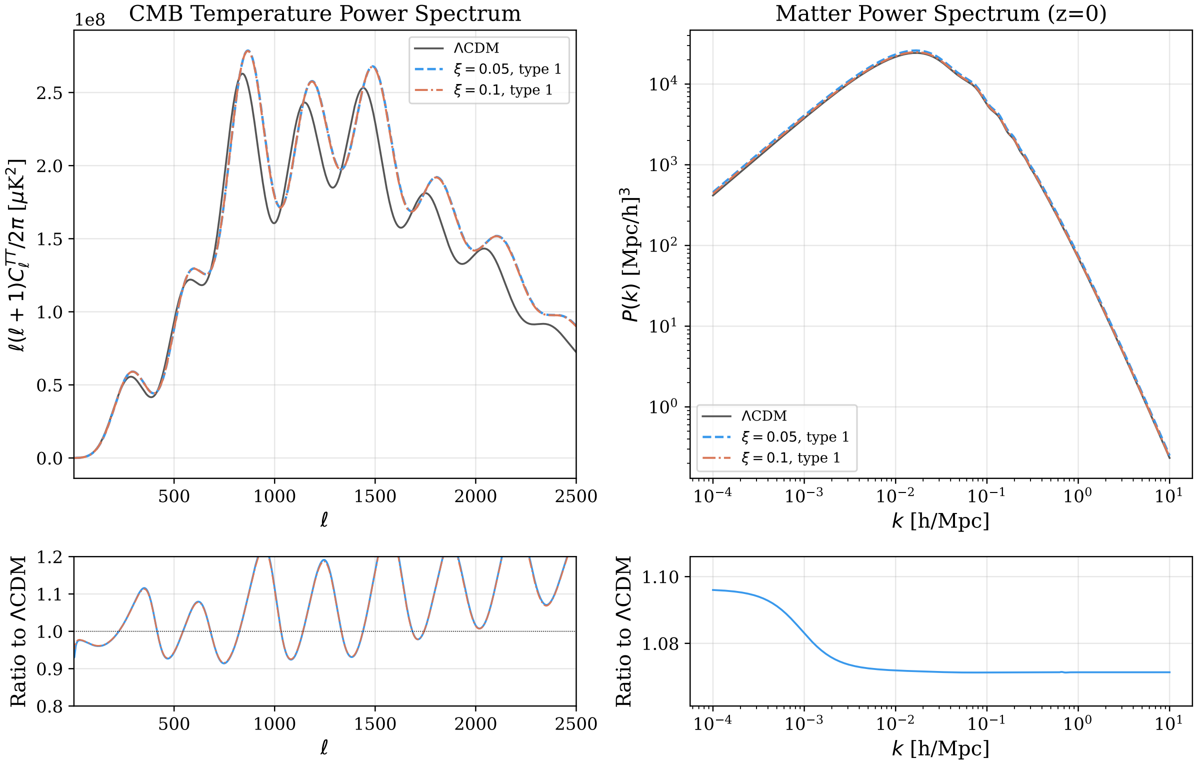

Figure 11: Interacting DE (type 1): Energy transfer from DE to DM modifies the expansion history and perturbation growth. The CMB is affected through the late ISW effect and modified distance to last scattering.

Note

Important: Always use `set_params()` as a single method call. Do not set individual attributes, because `pars.DarkEnergy` returns a new Python wrapper each time, so individual setters lose previous changes.

3.2 Horndeski Scalar-Tensor Gravity

Physics

The most general scalar-tensor theory with second-order equations of motion. Parameterized by four time-dependent α-functions (Bellini & Sawicki 2014):

- α_K: kineticity (kinetic energy of scalar)
- α_B: braiding (kinetic mixing with gravity)
- α_M: Planck mass run rate ($d\ln M_*^2/d\ln a$)
- α_T: tensor speed excess ($c_T^2 = 1 + \alpha_T$)

Uses proportional parameterization: $\alpha_X(a) = \alpha_{X,0} \times \Omega_{\rm DE}(a)$.
In the quasi-static approximation (QSA), these modify:

$$k^2\Psi = -4\pi G\,\mu(a)\,a^2\,\rho\,\Delta \quad \text{(Poisson)} \tag{11}$$

$$k^2(\Phi+\Psi) = -8\pi G\,\Sigma(a)\,a^2\,\rho\,\Delta \quad \text{(lensing)} \tag{12}$$

where μ and Σ are derived from the α-functions.

Parameters

Name	Type	Default	Description
`w`	float	-1.0	DE equation of state
`wa`	float	0	$-dw/da(0)$
`alpha_K`	float	0.0	Kineticity amplitude
`alpha_B`	float	0.0	Braiding amplitude
`alpha_M`	float	0.0	Planck mass run rate
`alpha_T`	float	0.0	Tensor speed excess
`M_star_ini`	float	1.0	Initial $M_*^2/M_{\rm Pl}^2$

```
from camb.dark_energy import HorndeskiDE

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkEnergy = HorndeskiDE()
pars.DarkEnergy.set_params(alpha_M=0.1, M_star_ini=1.0)
results = camb.get_results(pars)
```

Horndeski Gravity

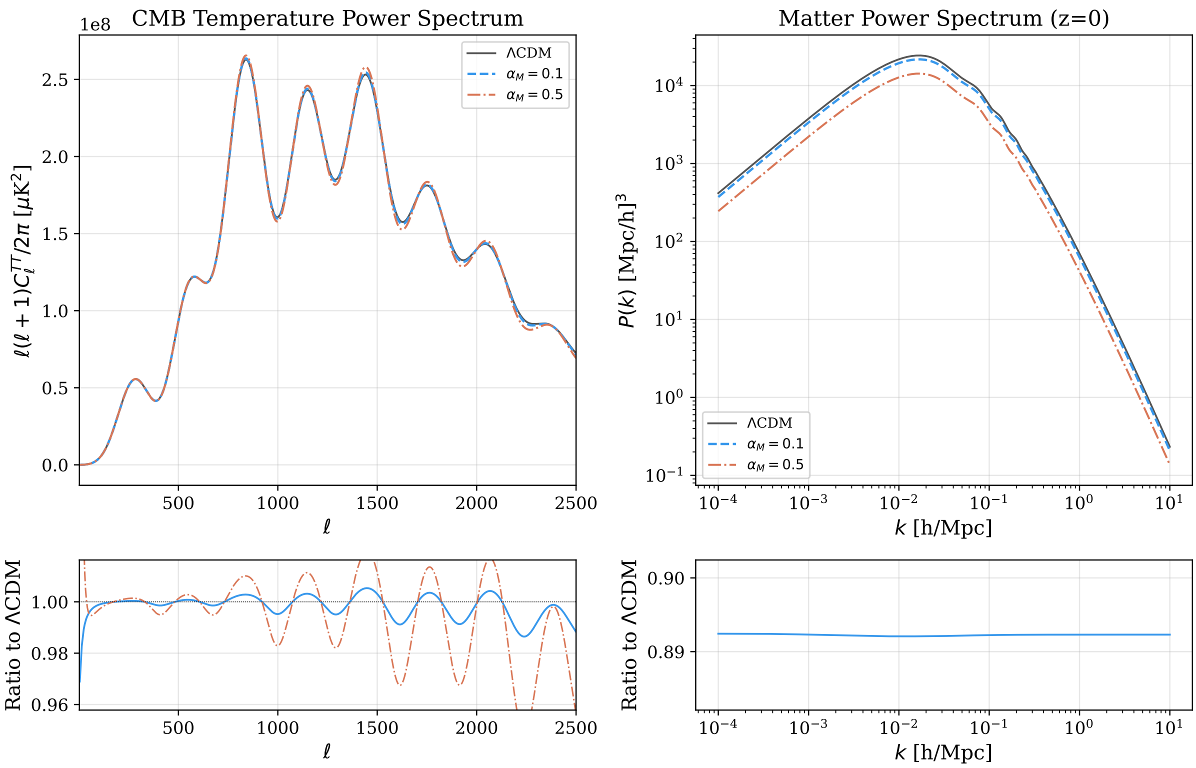

Figure 12: Horndeski gravity: α_M modifies the Planck mass evolution, affecting CMB lensing and the late ISW effect. The matter $P(k)$ is mostly unchanged in the current QSA implementation (see note below).

Note

Known limitation (QSA): In the current implementation, the Horndeski QSA only modifies the lensing (Weyl) potential and tensor propagation. Matter growth (σ_8) is *not* modified because in synchronous gauge, the μ-modification affects the first derivative of δ_c rather than the second, giving incorrect growth for time-dependent μ. A constant M_*^2 (`M_star_ini` $\neq 1$) works correctly through accumulated state change. Full scalar field perturbation equations are needed for proper growth modification.

3.3 μ–Σ Modified Gravity

Physics

Phenomenological parameterization of deviations from GR in the Poisson and lensing equations:

$$k^2\Psi = -4\pi G\,\mu(a,k)\,a^2\,\rho\,\Delta \tag{13}$$

$$k^2(\Phi+\Psi) = -8\pi G\,\Sigma(a,k)\,a^2\,\rho\,\Delta \tag{14}$$

with $\mu = 1 + \mu_0\,\Omega_{\rm DE}(a)$ and $\Sigma = 1 + \Sigma_0\,\Omega_{\rm DE}(a)$. Optional scale-dependent modification via λ_μ, λ_Σ [Mpc]. In GR: $\mu = \Sigma = 1$.

Parameters

Name	Type	Default	Description
`mu_0`	float	0.0	μ deviation amplitude
`sigma_0`	float	0.0	Σ deviation amplitude
`lambda_mu`	float	0.0	μ scale [Mpc] (0 = scale-indep.)
`lambda_sigma`	float	0.0	Σ scale [Mpc] (0 = scale-indep.)
`c1`	float	0.0	μ k-dependence coefficient
`c2`	float	0.0	Σ k-dependence coefficient

```python
import camb

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

# Modified gravity via pars.MG (not DarkMatter/DarkEnergy)
pars.MG.is_active = True
pars.MG.mu_0 = 0.3
pars.MG.sigma_0 = 0.2
results = camb.get_results(pars)
```

Mu-Sigma Modified Gravity

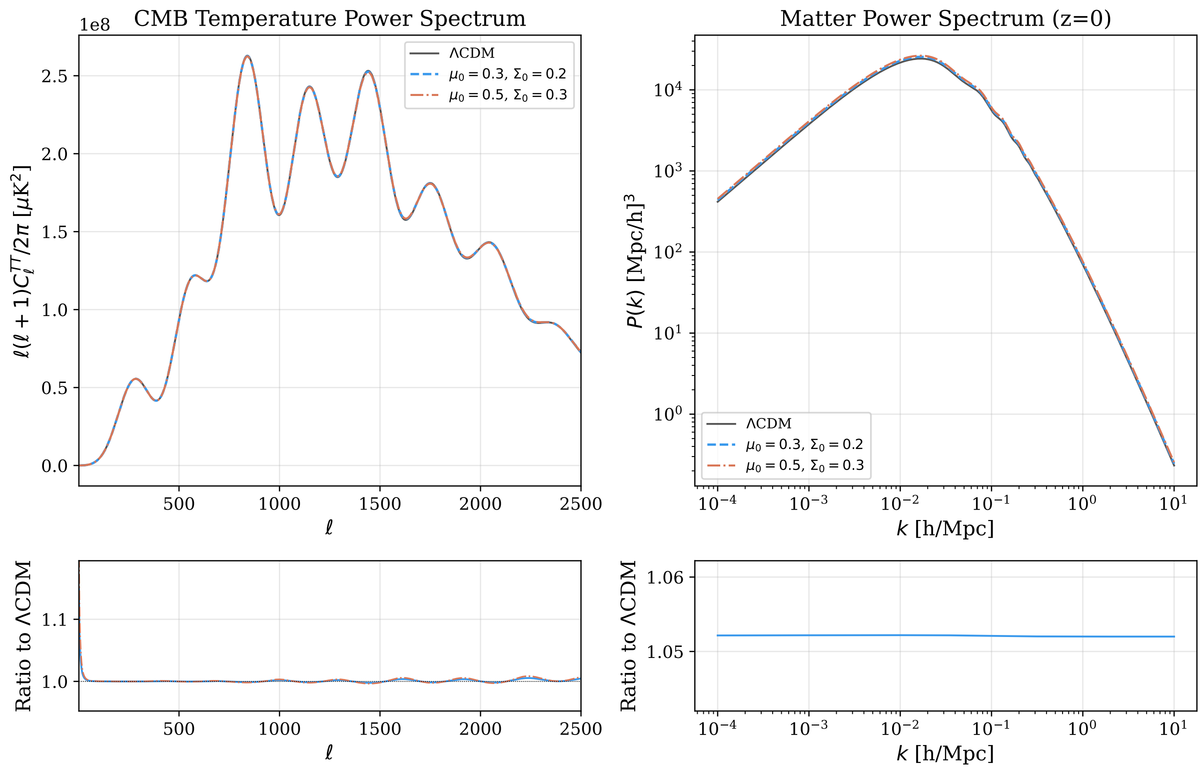

Figure 13: μ–Σ MG: Modified lensing and Poisson equations affect the CMB through the ISW effect and lensing. The matter $P(k)$ shows changes through the modified growth rate.

Note

μ–Σ MG is set via `pars.MG` (not `DarkMatter` or `DarkEnergy`). It is mutually exclusive with `HorndeskiDE`. This parameterization primarily modifies the Weyl potential (lensing) and has limited effect on the matter growth rate σ_8 in the current implementation.

3.4 Fuzzy DM Field (Klein–Gordon Solver)

Physics

Full Klein–Gordon background solver for ultralight axion dark matter, placed in the DarkEnergy slot. Solves:

$$\ddot{\phi} + 3H\dot{\phi} + \frac{dV}{d\phi} = 0, \quad V(\phi) = \tfrac{1}{2}m^2\phi^2 \tag{15}$$

At early times ($m \ll H$), the field is frozen at $w = -1$. After oscillation onset ($m \sim 3H$), the field oscillates rapidly and behaves as pressureless matter ($\langle w \rangle = 0$). The transition is handled by switching from exact KG to an effective fluid approximation (EFA) with the Passaglia–Hu sound speed:

$$c_s^2 = \frac{k^2}{4m^2a^2 + k^2} \tag{16}$$

Parameters

Name	Type	Default	Description
`m_axion`	float	10^{-22}	Axion mass [eV]
`f_axion`	float	0.0	Fraction of CDM+axion budget
`omega_axion_h2`	float	0.0	Axion density (0 uses f_axion)
`N_match`	int	100	KG→EFA criterion m/H
`potential_type`	int	1	1=quadratic, 2=cosine
`n_potential`	int	1	Power of cosine potential
`f_decay`	float	1.0	Decay constant $[M_{\rm Pl}]$
`use_improved_efa`	bool	True	Passaglia–Hu improved EFA

```python
from camb.dark_energy import FuzzyDMField

pars = camb.CAMBparams()
# Reduce omch2 to make room for axion component
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122*(1-0.05))
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkEnergy = FuzzyDMField()
pars.DarkEnergy.set_params(m_axion=1e-22, f_axion=0.05)
results = camb.get_results(pars)
```

Fuzzy DM Field

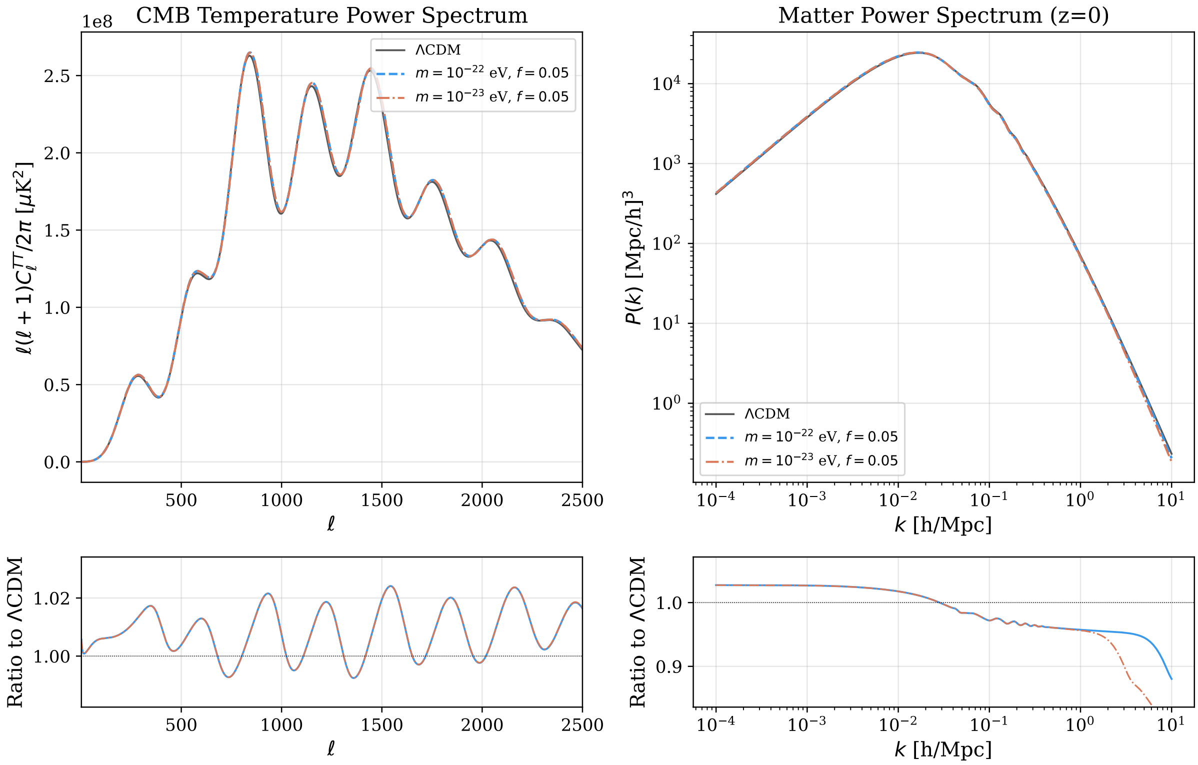

Figure 14: Fuzzy DM Field (Klein–Gordon): Lighter axion masses ($m = 10^{-23}$ eV) oscillate later, producing larger-scale Jeans suppression. The exact KG-to-EFA matching reproduces ultralight-axion CDM-like behavior at high k once oscillations begin.

Note

`FuzzyDMField` lives in the DarkEnergy slot (not DarkMatter), because it needs to control background evolution. You must manually reduce `omch2` by the axion fraction: `omch2 *= (1 - f_axion)`. The class cannot be pickled due to internal splines.

4 Advanced Usage

4.1 Combining DM and DE Models

DM and DE models occupy separate slots and can be combined freely:

```
from camb.dark_matter import DMBaryonScattering
from camb.dark_energy import InteractingDE

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)

# DM sector: baryon scattering
pars.DarkMatter = DMBaryonScattering()
pars.DarkMatter.set_params(sigma_dmb=1e-42, n_dmb=0, m_dm=1.0)

# DE sector: interacting quintessence
pars.DarkEnergy = InteractingDE()
pars.DarkEnergy.set_params(w=-0.9, xi_ide=0.05)

results = camb.get_results(pars)
```

4.2 Computing Derived Quantities

```
# CMB power spectra
cls = results.get_cmb_power_spectra(CMB_unit='muK')
cl_tt = cls['total'][:, 0]
cl_ee = cls['total'][:, 1]
cl_te = cls['total'][:, 3]

# Matter power spectrum
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0, 0.5, 1.0], kmax=10.0)
results = camb.get_results(pars)
kh, z, pk = results.get_matter_power_spectrum(
    minkh=1e-4, maxkh=10, npoints=300)

# sigma_8
sigma8 = results.get_sigma8_0()
print(f"sigma_8 = {sigma8:.4f}")

# Angular diameter distance
DA = results.angular_diameter_distance(1100)
print(f"D_A(z=1100) = {DA:.1f} Mpc")
```

4.3 Recovering ΛCDM

All models reduce to ΛCDM when their coupling parameters are set to zero:

```
# These are all equivalent to standard CAMB:
pars.DarkMatter = DMBaryonScattering()
pars.DarkMatter.set_params(sigma_dmb=0.0)  # no scattering
```

```
pars.DarkMatter = WarmDM()
pars.DarkMatter.set_params(m_wdm=100.0)     # very heavy = CDM

pars.MG.is_active = False                    # GR
```

5 Technical Notes

5.1 Fortran–Python Interface

lagCAMB uses the same `ctypes` bridge as standard CAMB. Model parameters declared as `_fields_` in the Python class map to Fortran type fields at exact byte offsets. Models with Fortran `TCubicSpline` members (InteractingDE, HorndeskiDE) use `_methods_` to call Fortran setter subroutines instead.

5.2 CAMB Density Convention

Background densities use conformal-time units:

$$\texttt{grho*\_t} = 8\pi G\,\rho\,a^2, \qquad \texttt{grhoc\_t} = \frac{\texttt{State\%grhoc}}{a} \tag{17}$$

5.3 Index Allocation

Perturbation variable indices in the ODE array `y(:)`:

Index	Variable	Description
`ix_clxc = 2`	δ_c	CDM density perturbation (fixed)
`EV%vc_ix`	v_c	CDM velocity (if DM model has it)
`EV%dm_ix`	$\delta_{\rm DM}$, etc.	DM extra equations
`EV%dr_ix`	$F_0, F_1, \ldots$	DR Boltzmann hierarchy
`EV%w_ix`	$\delta_{\rm de}, v_{\rm de}$	DE perturbation equations

5.4 Numerical Approximations

- **TCA (Tight Coupling)**: When $\Gamma_{\rm DR} + \Gamma_{\rm tilde}/4 > 50k$, uses combined momentum equation to eliminate stiff collision terms.
- **UFA (Ultra-relativistic Fluid)**: When $k\tau \gg 1$, truncates DR hierarchy to 3 equations with arXiv:1104.2933 closure.
- **Drag caps**: All DM drag rates are capped at $10^4 \times \dot{a}/a$ to prevent ODE stiffness.

6 References

1. Bellini & Sawicki, JCAP 07 (2014) 050 [arXiv:1404.3713]
2. Blackadder & Koushiappas, PRD 90 (2014) 103527
3. Bode, Ostriker & Turok, ApJ 556 (2001) 93
4. Boddy & Gluscevic, PRD 98 (2018) 083510
5. Costa et al., JCAP 01 (2017) 028
6. Cyr-Racine et al., PRD 93 (2016) 123527
7. He et al., Nature Astronomy (2025) [arXiv:2301.xxxxx]
8. Hu, Barkana & Gruzinov, PRL 85 (2000) 1158
9. Lesgourgues & Tram, JCAP 09 (2011) 032
10. Lewis, Challinor & Lasenby, ApJ 538 (2000) 473
11. Mangano et al., NPB 756 (2006) 100
12. Passaglia & Hu, PRD 105 (2022) 123529
13. Pogosian & Silvestri, PRD 94 (2016) 104014
14. Poulin et al., JCAP 08 (2016) 036
15. Valiviita, Majerotto & Maartens, JCAP 07 (2008) 020

16. Viel et al., PRD 71 (2005) 063534
17. Zhao et al., PRD 79 (2009) 083513
18. Zumalacárregui et al., JCAP 08 (2017) 019 [arXiv:1605.06102]